Based on the topics in Session 4: Deep Parsing and Tools for NLP, the following laboratory programs are organized to progress from parsing fundamentals to semantic and discourse-level applications.

NLP Laboratory Programs – Session 4

Exp. No.	Laboratory Program
1	Implement a Context-Free Grammar (CFG) parser using NLTK and generate parse trees for given sentences.
2	Develop a Recursive Descent Parser to validate sentence structures based on predefined grammar rules.
3	Implement Dependency Parsing using spaCy and visualize dependency relationships among words.
4	Implement a Context-Free Grammar (CFG) using NLTK, parse given sentences, and generate constituency parse trees to represent their grammatical structure.
5	Compare Constituency Parsing and Dependency Parsing on a set of sample sentences and analyze outputs.
6	Implement the CYK parsing algorithm to check whether a sentence belongs to a given grammar. Syntactic Parsing using CYK algorithm
7	Implement Semantic Role Labeling (SRL) using Natural Language Processing techniques to identify predicates and classify their arguments into semantic roles such as Agent, Action, Patient, and Instrument. Analyze the output for a set of sample sentences.
8	Implement a Semantic Parsing system using Natural Language Processing techniques to convert simple natural language sentences into logical representations using grammar rules. Analyze the generated logical representations for a set of sample sentences.
9	Implement an Information Extraction system using Named Entity Recognition (NER) with spaCy to identify and extract named entities such as persons, organizations, locations, dates, and other entity types from a given text. Analyze the extracted entities for a set of sample sentences.
10	Implement a Relation Extraction system using NLP techniques to identify and extract relationships between named entities from a given text. Extract relations such as Person–Organization and Organization–Location using suitable linguistic patterns, and analyze the extracted relationships for a set of sample sentences.
11	Implement an Automatic Text Summarization system using NLP techniques to generate extractive summaries from a given text. Apply sentence scoring and the TextRank algorithm to identify and rank important sentences, select the top-ranked sentences, and analyze the generated summaries for a set of sample documents.
12	Implement an Anaphora and Coreference Resolution system using NLP techniques to identify pronouns and determine their corresponding antecedents in a given text. Apply coreference resolution techniques to establish relationships between referring expressions and the entities they refer to, and analyze the results for a set of sample sentences.
13	Pragmatic Analysis of Natural Language. Analyze sentences to identify speech acts, intentions and contextual meaning.
14	Discourse Analysis and Coherence. Analyze relationships between sentences using discourse markers, sentence similarity and coherence.
15	Ontology Creation and Semantic Web using RDF. Create a simple ontology, represent knowledge using RDF triples and query it using SPARQL

Course Outcome Mapping

Beginner Level	CFG Parsing using NLTK Recursive Descent Parser Dependency Parsing Constituency Parsing CYK Parsing

Intermediate Level	Semantic Role Labeling Semantic Parsing Named Entity Recognition Relation Extraction Text Summarization Anaphora and Coreference Resolution
Advanced Level	Pragmatic Analysis Discourse Analysis and Coherence Ontology Creation RDF and SPARQL Querying

Suggested Software Tools

- Python 3.x
- NLTK
- spaCy
- Stanford CoreNLP
- Gensim
- Hugging Face Transformers
- Protégé
- RDFLib / Apache Jena